Distinguish a JMF target, production variability and the actual contractual acceptance criteria. Quality-control results help manage the process, while acceptance/audit results serve their specified role. Retain original results, corrections and disposition records.

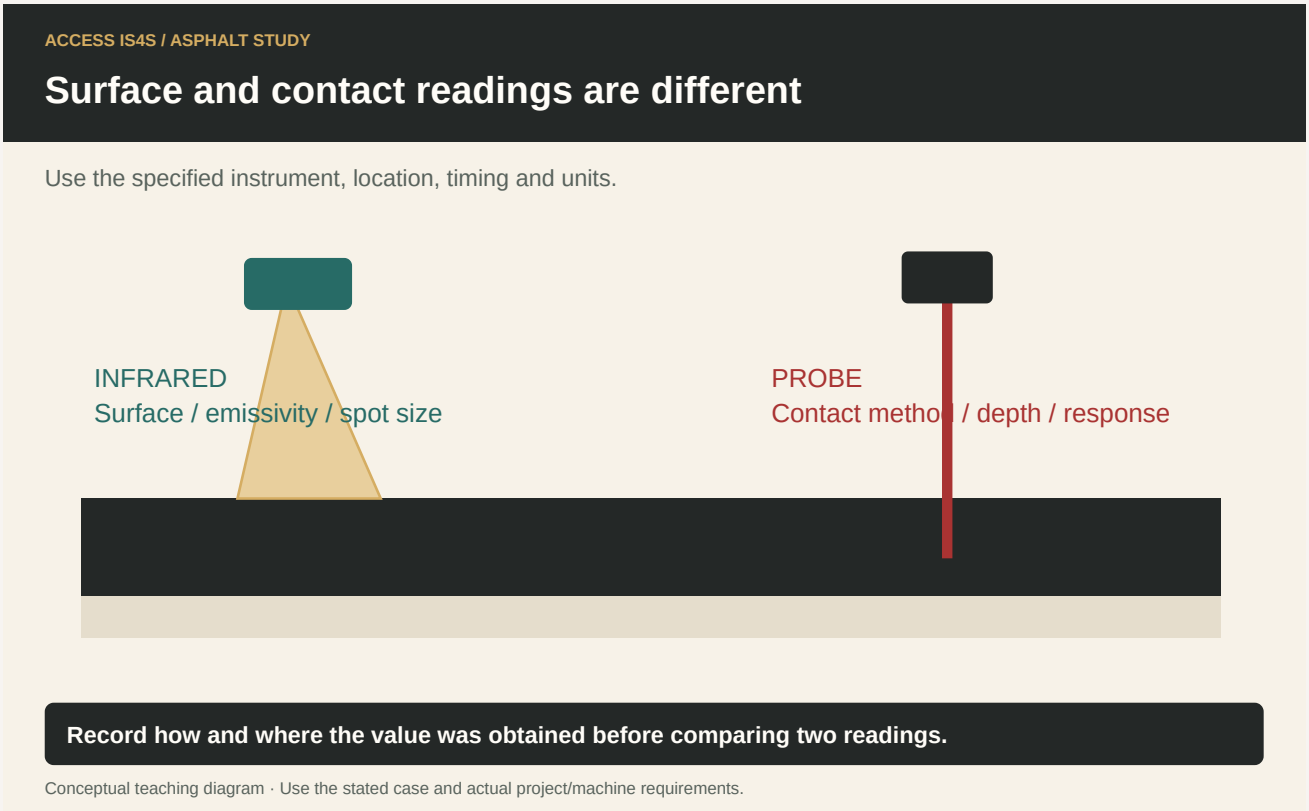
LEARNING STEP 4

The process of taking mix temperature readings

Select the temperature instrument and method for the required measurement. An infrared device senses a surface within its field of view; spot size, viewing conditions and surface properties affect the reading. A probe measures at its specified contact/insertion location. Neither automatically represents the entire load.

Verify instrument identity, condition and required check/calibration status. Record the time, location, product/load, units and method. If readings disagree, investigate the different measurement conditions before averaging them. A Celsius/Fahrenheit conversion changes units, not where the temperature was measured.

Surface and contact readings are different



Note. ACCESS IS4S. (2026). Surface and contact readings are different [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

Worked example · A result with a broken chain

Fictional bag label says aggregate bin A, 09:15. The sampling log says bin B, 09:15, and the test sheet uses the A label. The test values are within the limits printed on the teaching card.

Think it through.

The identity conflict prevents a reliable match to the intended material. Preserve all records, hold the interpretation and resolve the sampling chain through the responsible quality role. Passing numbers do not repair identity. If the source cannot be established, the required resampling/disposition process must be used.

Try it and explain your thinking

Build a sample route from planned location to test report

Build a sample route from planned location to test report. Identify where representativeness, safety and identity can each fail.

Skills you use: Reading, Problem solving, Communication.

Convert a fictional recorded 140°C to Fahrenheit using F = 1.8C + 32

Convert a fictional recorded 140°C to Fahrenheit using F = 1.8C + 32. Explain why a 140°C surface reading is not proof of the internal temperature.

Skills you use: Numeracy, Writing.

Save a sample register for the worked case

Save a sample register for the worked case. A partner supplies a corrected log with an author/date; retain the original and document the resolution rather than overwriting history.

Skills you use: Digital, Collaboration, Adaptability, Creativity and innovation.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Use the actual approved sampling and hazardous-energy/access procedures with qualified supervision.	0 / 1 / 2	Evidence/help: _____
Observe representative collection/reduction, contamination prevention, labelling and temperature method/recording.	0 / 1 / 2	Evidence/help: _____
Do not stage live-belt access or hot-material handling without the complete authorized arrangement.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, E2, printed p. 45; source PDF pp. 47–47.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 10: Quality Control and Acceptance of Hot Mix Asphalt; source PDF pp. 479–552. Open publication.
- BC Ministry of Transportation and Transit. (2025). 2025 Standard Specifications for Highway Construction, Volume 2. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Study chapter · Outline E3

Determine haul truck samples

A truck sample connects the delivered mixture to a production and quality record. The truck, load ticket, mix identifier, time and sampled location must remain traceable. Sampling should answer the planned question; choosing only the easiest or coolest-looking portion can introduce bias. Safe access is established before sampling, not improvised after a truck arrives.

- Mix sampling
- Quality control
- Bag samples

LEARNING STEP 1

Mix sampling

Confirm the intended material and sampling purpose: mixture identity, temperature, binder-content/gradation testing or another specified property. Use the required sampling location, increments, tools and sample size under the approved method. The arrangement must control vehicle movement, hot material, access and other exposed people. Do not climb a load or reach into a discharge stream on the strength of a classroom description.

Temperature and laboratory specimens may require different handling. The sampling record identifies exactly what was taken, where, when and by whom. A result is compared with the correct formula and contractual criterion, not simply with the last truck’s appearance.

A selected load keeps its identity



Note. ACCESS IS4S. (2026). A selected load keeps its identity [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

LEARNING STEP 2

Quality control

Random selection reduces the opportunity to choose favourable loads or locations, while the approved plan determines the sampling population and timing. Random does not mean unplanned or unsafe. The testing agency/company personnel must have the required competence and authorization, and the truck/paver crew must understand the hold and communication arrangement.

Keep quality-control sampling separate from owner acceptance/audit sampling where the contract distinguishes them. A failed result is handled through the documented process, preserving the selected sample and records. Replacing it with an unrecorded convenient sample defeats traceability and can hide a production problem.

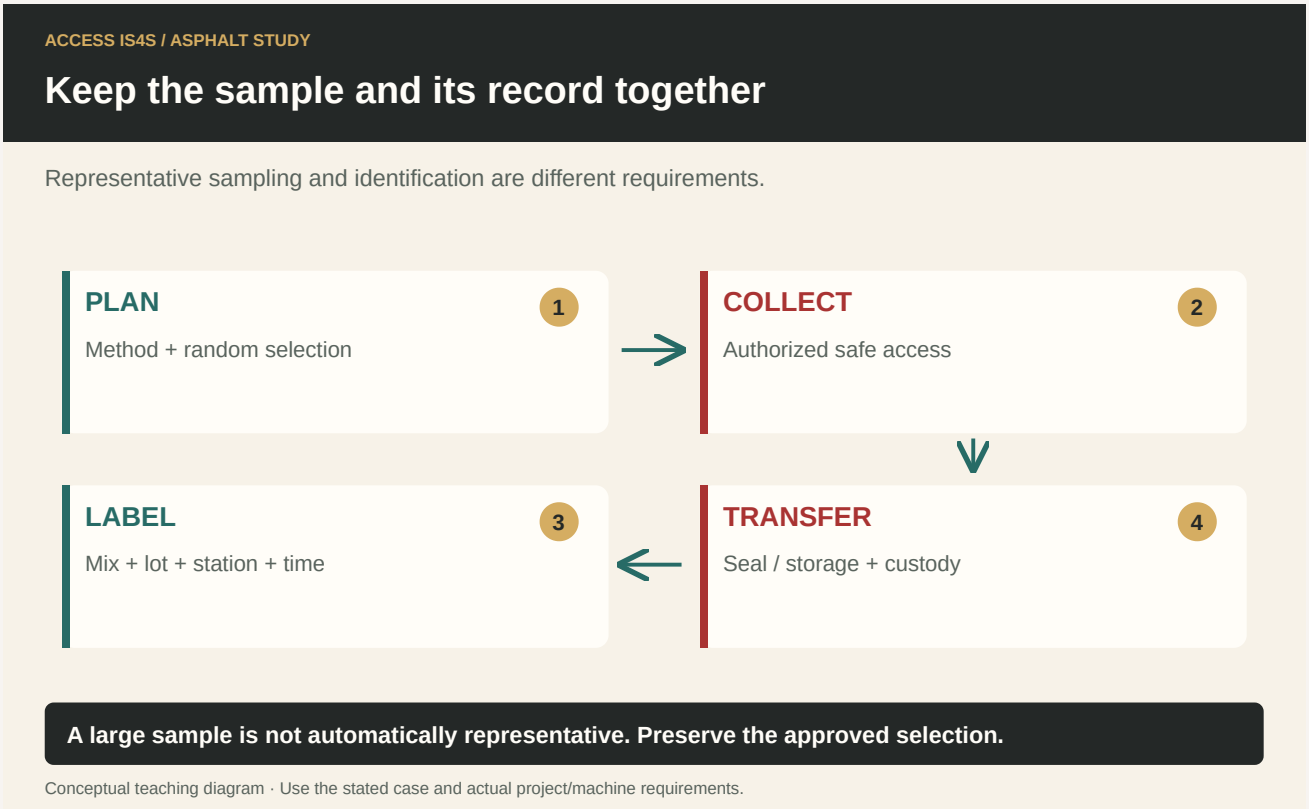
LEARNING STEP 3

Bag samples

Use the container or bag specified for the sample's material, temperature, quantity and test. Identify the load, mix, date/time, location, sampler, intended test and any required seal/backup identification. Protect the sample from contamination and loss while following safe hot-material handling. An ordinary classroom bag may be unsuitable for actual hot mix.

A backup sample is useful only if it remains correctly identified, preserved and handled under the applicable method and retention arrangement. Record transfer and receipt. If a seal breaks or the label becomes unreadable, report the condition rather than recreating identity from memory.

Keep the sample and its record together



Note. ACCESS IS4S. (2026). Keep the sample and its record together [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

Worked example · Random selection with traceability

Fictional teaching plan identifies loads 1–10 and a recorded random selection of loads 3 and 8. Load 8 is delayed; load 7 is easier to reach. The sampler proposes silently substituting 7.

Think it through.

The approved selection has changed. Follow the plan’s authorized contingency/rescheduling method and record any change; do not silently substitute a convenient load. Maintain safe coordination while waiting or arranging the approved alternative. Random selection and safe access must both be satisfied.

Try it and explain your thinking

Explain and apply the idea

Prepare the complete label and transfer record for fictional load 3, mix JMF-C revision 2, sampled at 10:25 for binder/gradation testing. Mark any information the case does not supply.

Skills you use: Reading, Writing.

Two fictional increments are 6.2 kg and 5.8 kg

Two fictional increments are 6.2 kg and 5.8 kg. Find total mass and explain why this sum alone cannot establish a representative or correctly reduced laboratory sample.

Skills you use: Numeracy, Problem solving, Communication.

Communicate and verify the work

Create a digital sampling log, introduce the delayed-load change with a partner and preserve the authorized decision trail.

Skills you use: Digital, Collaboration, Adaptability, Creativity and innovation.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Instructor establishes actual vehicle separation, access, tools, containers and sampling method with the competent testing role.	0 / 1 / 2	Evidence/help: _____
Observe representative sampling from authorized locations, identification, safe handling and backup/transfer procedure.	0 / 1 / 2	Evidence/help: _____
Record practical evidence from the actual task; paper labels alone do not demonstrate hot-mix sampling skill.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, E3, printed p. 46; source PDF pp. 48–48.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 10: Quality Control and Acceptance of Hot Mix Asphalt; source PDF pp. 479–552. Open publication.
- BC Ministry of Transportation and Transit. (2025). 2025 Standard Specifications for Highway Construction, Volume 2. Open publication.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 16: Mobile Equipment. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Determine compaction levels

Density relates mass to volume. Compaction changes the arrangement of particles and air voids, but the measured density must be compared with the correct reference and method. Two instruments can report numbers that look similar while representing different locations or measurement principles. Keep the instrument, sample, units and denominator visible.

Two methods to determine compaction

LEARNING STEP 1

Two methods to determine compaction

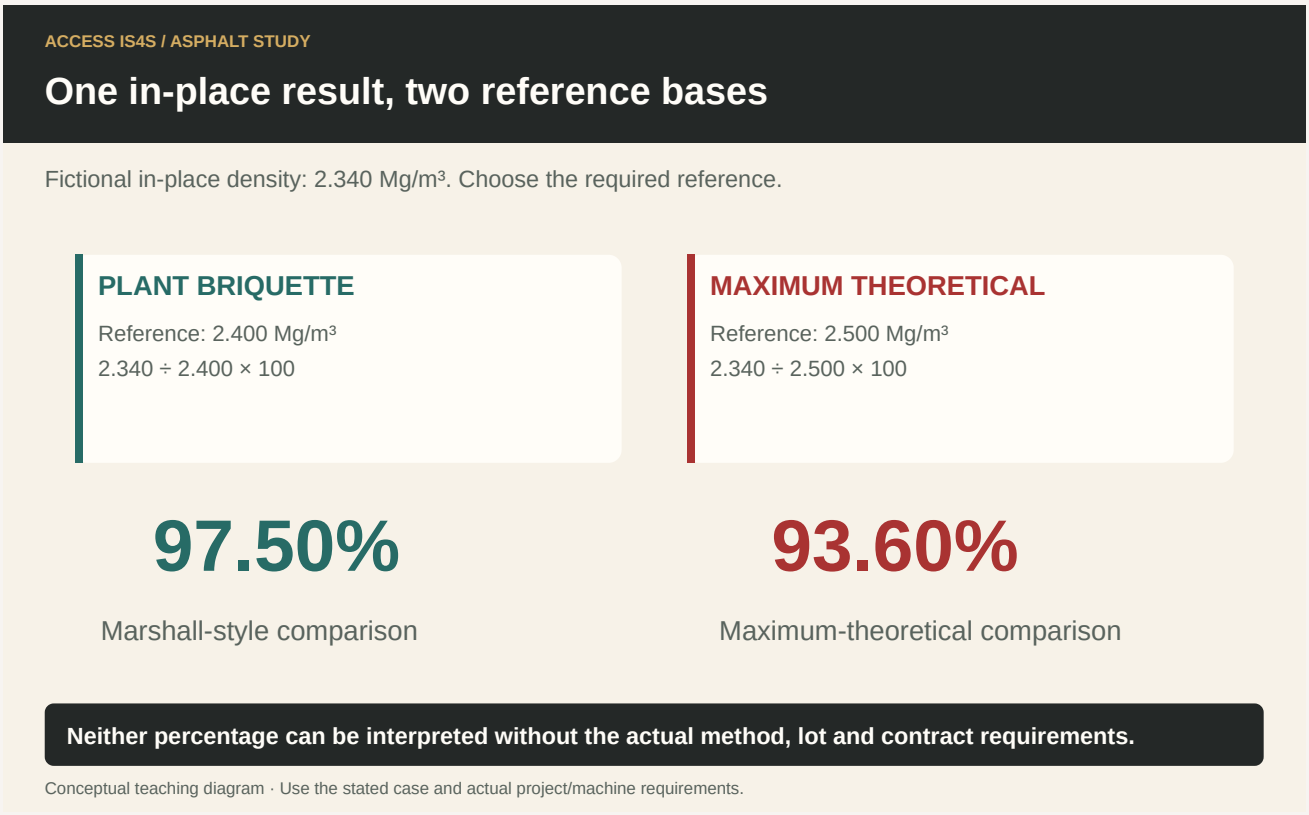
A **nuclear density gauge** uses a radioactive source and detector arrangement to infer a material property through its calibrated method. The actual instrument, mode, calibration/standardization, surface preparation and test method affect the result. In Canada, possession/use and transport require the applicable CNSC licensing, training, security and radiation-safety arrangements. Reading the Troxler 3430 manual is useful instrument study; it does not authorize source handling or operation.

A **core** is a physical specimen removed at an identified location. The approved coring and laboratory method establish sampling, handling, dimensions and density determination. Coring also requires utility/location checks, traffic/access controls and safe equipment/water management, followed by the specified repair of the hole. A damaged, contaminated or mislabelled core may not support the intended result.

Gauge and core results are compared only through the specified procedure and any required correlation. A percentage density must state its reference: maximum theoretical mixture density, laboratory compacted reference or another defined basis. Do not compare percentages that use different denominators. The result belongs to the sampled location and acceptance unit defined by the contract; one point does not automatically represent an entire project.

Report original readings, checks, location and any limitation. A low result prompts the approved investigation/disposition process. Additional rolling is not automatically appropriate after the material's workable window or when the cause is unknown.

One in-place result, two reference bases



Note. ACCESS IS4S. (2026). One in-place result, two reference bases [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

Troxler 3430 gauge and accessories

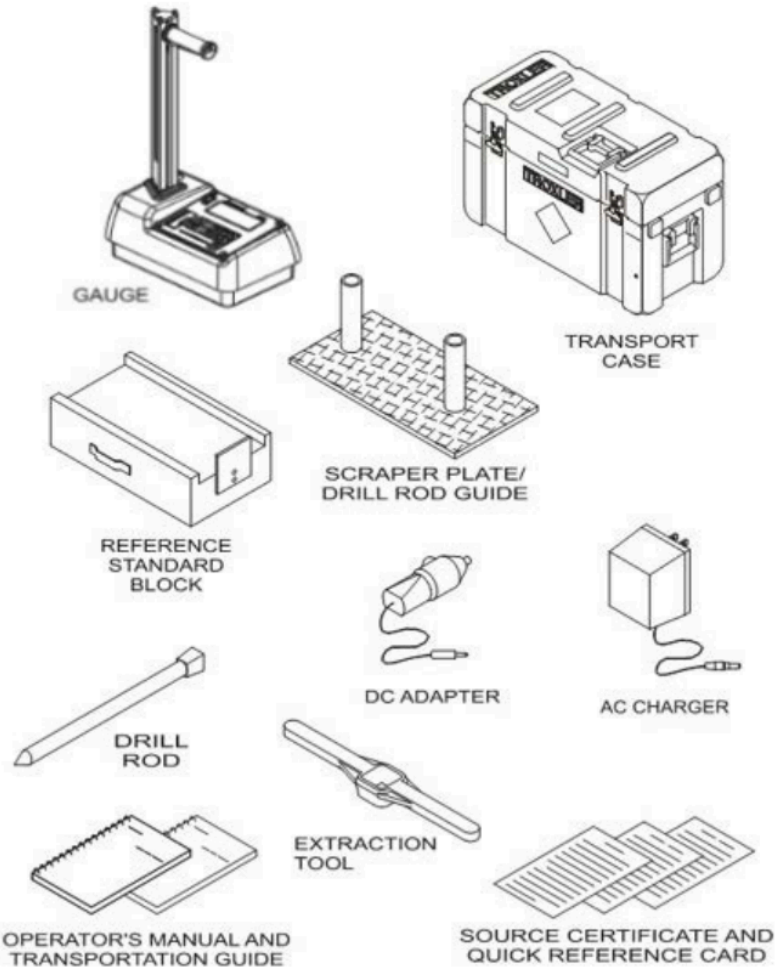


Figure 1. Model 3430 Gauge and Accessories

Note. Troxler Electronic Laboratories. (2021). RoadReader Model 3430: Manual of operation and instruction (Ed. 1.3, Fig. 1, p. 1-5; PDF p. 19). Identification only; Canadian use requires applicable CNSC authorization. Full source context.

Worked example · The denominator changes the meaning

Fictional in-place density is 2.34 t/m³. Reference A, maximum theoretical density, is 2.50 t/m³. Reference B, a compacted laboratory reference in a different teaching system, is 2.40 t/m³.

Think it through.

Against A: 2.34/2.50 × 100 = 93.6%. Against B: 2.34/2.40 × 100 = 97.5%. Both calculations use the same numerator; they answer different defined questions. Neither establishes acceptance without the applicable method and criterion. Never relabel 97.5% as percent of maximum theoretical density.

Try it and explain your thinking

Explain and apply the idea

Compare gauge and core methods using columns for measurement principle, identification, checks, access and authorization.

Skills you use: Reading, Writing, Problem solving.

Repeat the worked ratios for an in-place value of 2.30 t/m³

Repeat the worked ratios for an in-place value of 2.30 t/m³. Write each full label.

Skills you use: Numeracy, Communication.

Communicate and verify the work

Build a saved density-result sheet and peer-test whether its denominator, units and test location can be found immediately. Correct any ambiguity.

Skills you use: Digital, Collaboration, Adaptability, Creativity and innovation.

Check your understanding of density evidence

Five questions, four marks each: 20 marks total. Minimum 14/20 (70%), as required by the outline for this knowledge check. This does not authorize gauge operation.

Question 1 · Explain one difference between core testing and nuclear-gauge testing, and identify the Canadian authorization issue.

Question 2 · Calculate percent density for 2.30 Mg/m³ in place and 2.50 Mg/m³ maximum theoretical.

Question 3 · Name four traceability fields needed with a density result.

Question 4 · A draft changes the reference density to improve the percentage. Explain the response.

Question 5 · Plan a safe learning observation of core or gauge testing.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Use a licensed/authorized gauge arrangement or approved core-sampling/laboratory arrangement with qualified supervision.	0 / 1 / 2	Evidence/help: _____
Assess method recognition, traceability, calculation and correct interpretation; no learner handles a radiation source without required authorization.	0 / 1 / 2	Evidence/help: _____

What the observer checks	Result	Evidence
For the outline knowledge quiz, use the questions in this unit with the provider's grading record; practical testing remains a separately observed outcome.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, E4, printed p. 47; source PDF pp. 49–49.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 9: Hot Mix Asphalt Compaction; source PDF pp. 447–478. Open publication.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 10: Quality Control and Acceptance of Hot Mix Asphalt; source PDF pp. 479–552. Open publication.
- Troxler Electronic Laboratories. (Edition 1.3, February 2021). RoadReader Model 3430 Surface Moisture-Density Gauge: Manual of Operation and Instruction. Open publication.
- Canadian Nuclear Safety Commission. (Web snapshot: October 30, 2026). Portable Nuclear Gauges. Open publication.
- Canadian Nuclear Safety Commission. (n.d.; current publisher download). Portable Gauge Quick Reference Guide. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Integrated practice · all nine Skills for Success

Trace a material and its sample

> This is a complete fictional classroom case. It does not describe an actual community, Nation, employer or project. Learners may use the supplied case without disclosing personal or cultural knowledge.

A fictional 1,200 g aggregate sample has retained masses of 180 g, 420 g, 360 g and 240 g in the pan. Its log says Mix M, lot 4, but the container label says lot 5. A draft report says “the sample is representative because it is large” and “20% RAP means 20% lower emissions.” The project location is in Metro Vancouver. No acceptance limits or permit number are supplied.

Reading · Check the facts

Identify the controlling facts, missing information and one unsupported statement in the brief.

Writing · Make the handover useful

Draft a concise handover with location, quantities, status and the next responsible action.

Numeracy · Calculate and check

Show the case calculation with units and an independent reasonableness check.

Digital · Direct AI and verify the files

Use an available AI assistant to draft a findings table and editable record from this brief. Request DOCX or a structured document, XLSX with formulas where useful, and a PDF/HTML reading copy. Audit every claim and reopen the actual outputs.

Problem solving · Investigate the discrepancy

Explain the discrepancy, what evidence is needed and who should authorize the next step.

Communication · Send and check the message

Brief a partner, ask for a check-back and repair one misunderstanding if it occurs.

Collaboration · Review together

Agree reviewer/author roles, complete a review and record each person’s contribution.

Adaptability · Respond to the change

Apply the change card, identify what changes and what stays, and revise the record.

Creativity and innovation · Make understanding easier

Create a labelled SVG, annotated plan, short audio explanation or captioned storyboard that makes the issue easier to understand. Test it with a reader/listener and improve one useful detail.

Change card · Use after the first attempt

The quality lead supplies a corrected source record linking the container to lot 5, with documented authorization and preserved original record.

Individual evidence record

Keep the brief, calculation, saved work, review findings and revision. Record each person's actual contribution and support. Use Not observed, Not yet demonstrated, Demonstrated with support, or Demonstrated independently for the stated action; these are local feedback descriptions.

For AI/media work, a transcript, script or storyboard is distinct from a generated audio/video file. Name what was actually produced and inspected. Do not upload private records or community-held material without the relevant authority.